

# Investigating the Effectiveness of Ashwagandha in Decreasing Levels of Nitric Oxide Induced by Neuroinflammation with LPS in SH-SY5Y Cells

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## Background

- Inflammation** → Immune response anywhere in the body
- Neuroinflammation** → Inflammation in the central nervous system (CNS), involving microglia and astrocytes (Soufli et al. 2016)

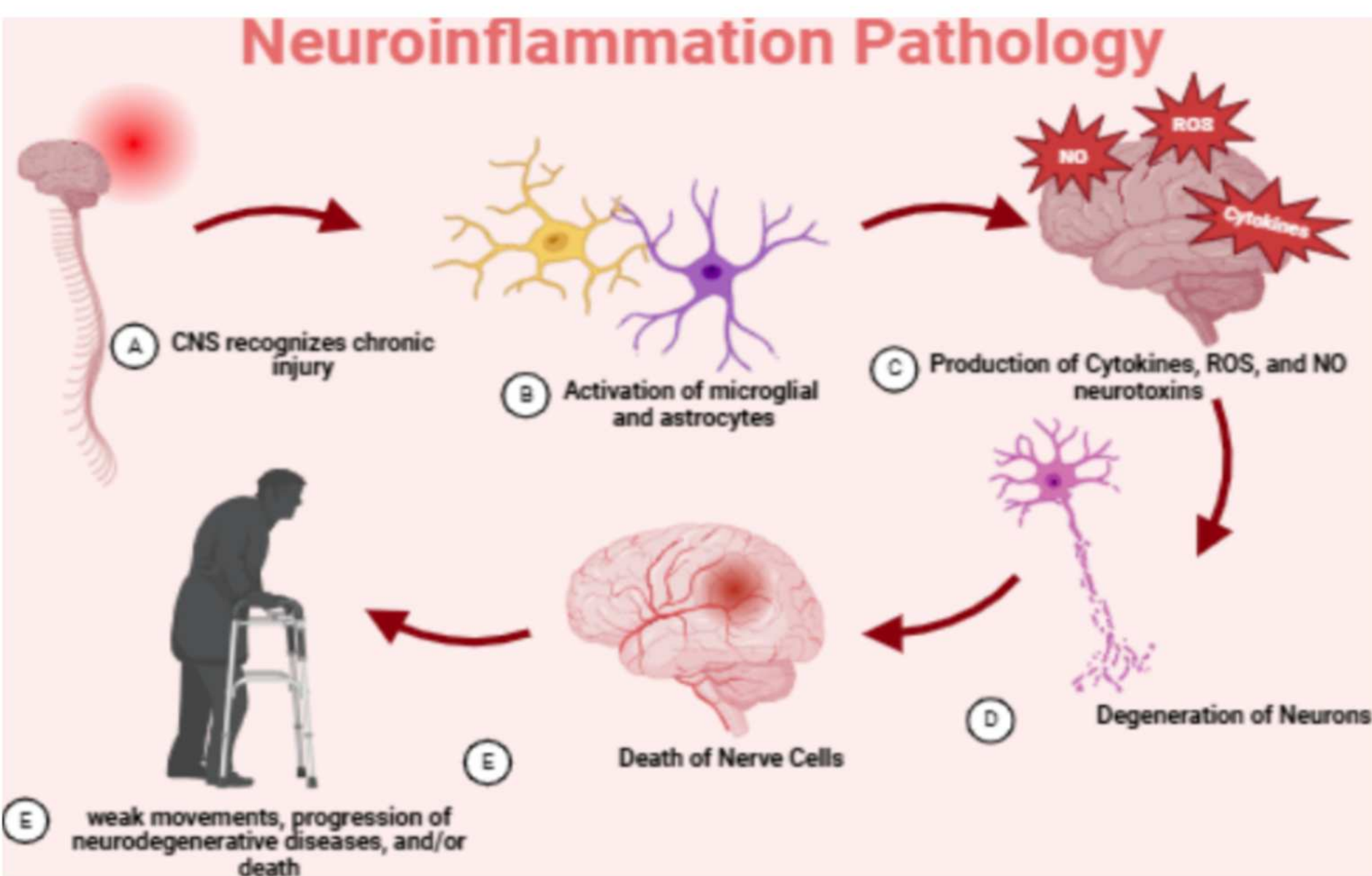


Figure 1. Image of the pathology of neuroinflammation

- Nitric oxide and proinflammatory cytokine production have a direct relationship.**



### Ashwagandha (Treatment):

- contains bioactive compounds (Withaferin A & Withanolide A) that decrease inflammation (Nile et al. 2021)

- LPS** → triggers inflammation by mimicking a bacterial infection, causing the release of proinflammatory cytokines and nitric oxide.

## Purpose & Hypothesis

**Purpose:** To determine whether Ashwagandha has the potential to decrease levels of nitric oxide (preventing the development of neurodegenerative diseases).

**Hypothesis:** If a diluted concentration of Ashwagandha is added to neuronal cells undergoing neuroinflammation, then nitric oxide levels will decrease due to the anti-inflammatory effects of Ashwagandha's bioactive compounds.

## Method

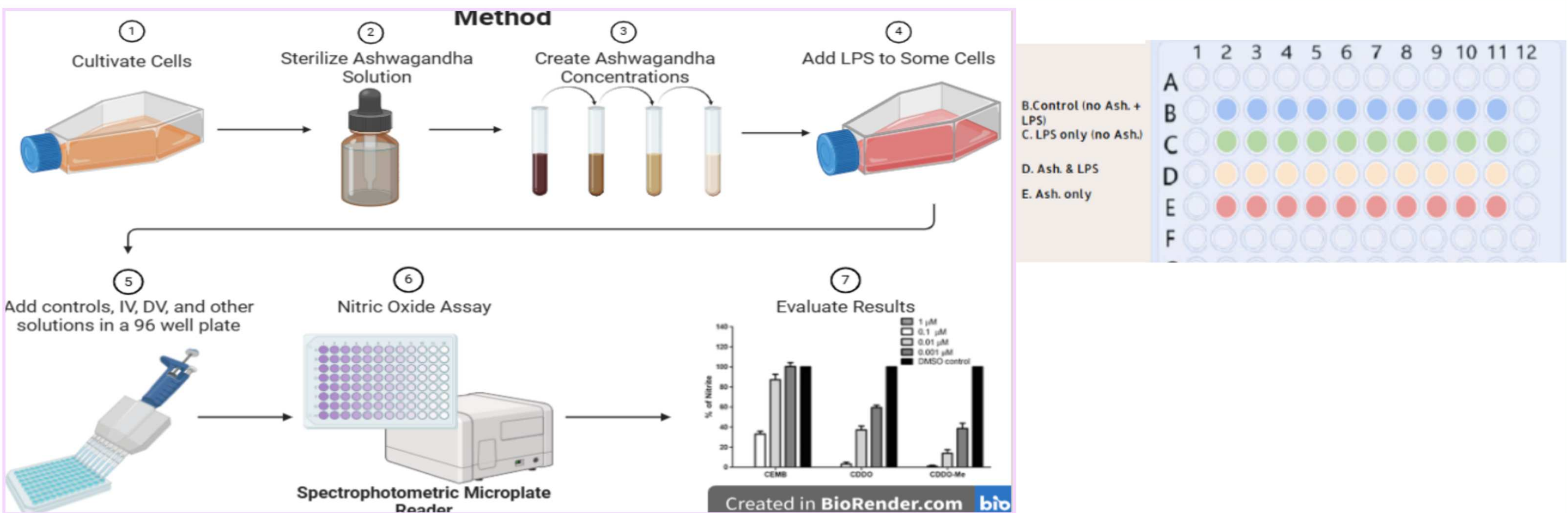


Figure 3. Illustration of the overview displaying the method used to collect nitric oxide levels of each experimental group.

### Variables:

**IV** - Presence of Ashwagandha Extract (1% of Ash. stock)

**DV** - Absorbance Levels (Nitric Oxide Levels)

**Constants** - Materials, temperature and duration of incubation, and concentration of solute & solvent (LPS concentration is 0.1 µg/mL)  
**Control**- Untreated SH-SY5Y cells

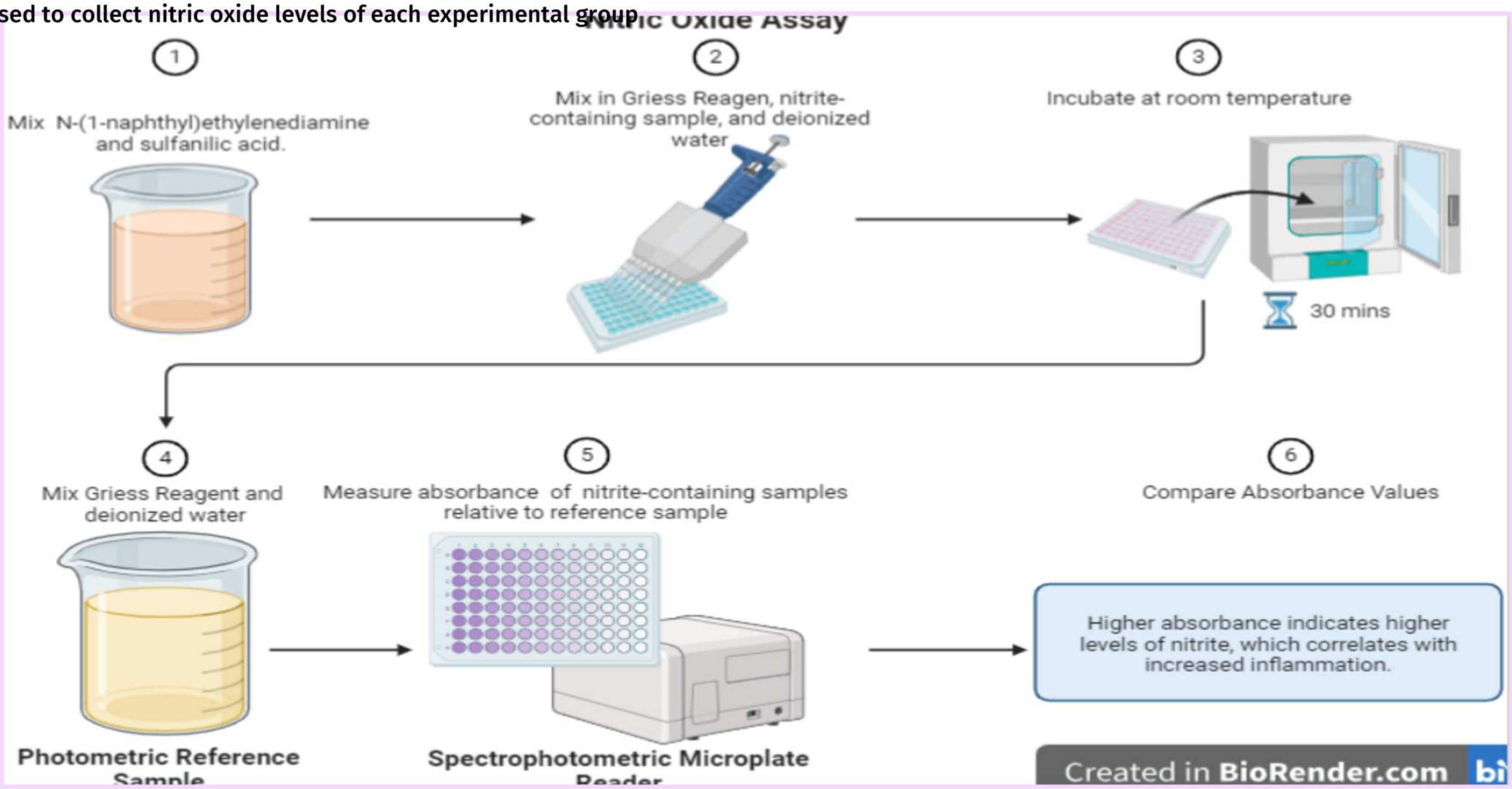
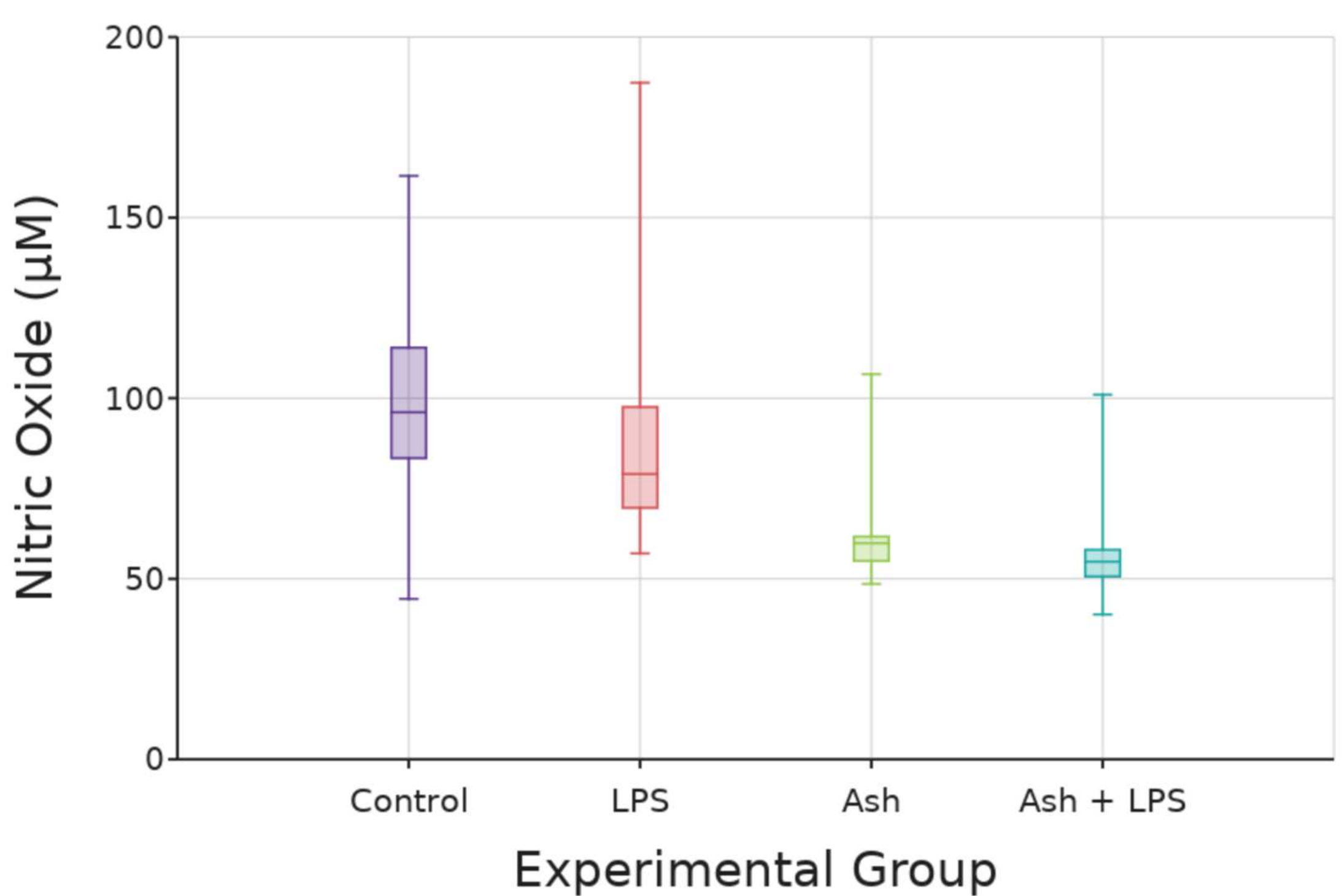


Figure 4. Illustration of Nitric Oxide Assay Steps

## Results

### Effect of Ashwagandha on Nitric Oxide Production in SH-SY5Y Cells Induced with Neuroinflammation



Note: Box and whiskers plot shows the median value (line), interquartile range (box), and the extent of the data (whiskers above and below at min / max data points).

### Graph Driven Test:

Due to the small number of replicates in each group, a Kruskal-Wallis test followed by Dunn's post hoc test was performed to determine whether there were significant differences in nitric oxide levels among the treatment groups.

| Degrees of Freedom(df) | H-statistic | P-value | Interpretation of P                                                                 |
|------------------------|-------------|---------|-------------------------------------------------------------------------------------|
| 3                      | 61.04       | <0.01   | A P-value of <0.01 means extremely strong confidence that the groups are different. |

Figure 5. Degrees of freedom, H-statistic, and P-value calculated from the Kruskal Wallis Test comparing the effect of Ashwagandha on nitric oxide levels.

## Discussion

The Kruskal-Wallis test revealed a statistically significant difference in nitric oxide production across the treatment groups ( $p < 0.01$ ).

The Dunn's post hoc analysis revealed that nitric oxide (NO) levels were significantly lower in the Ash and Ash + LPS groups compared to Control and LPS groups ( $p < 0.01$ ). There was no difference between Control and LPS ( $p = 1.00$ ), suggesting LPS alone did not increase NO. Likewise, no difference was found between Ash and Ash + LPS ( $p = 0.89$ ), indicating that Ashwagandha reduced NO levels even in the presence of inflammation. These results support Ashwagandha's potential anti-inflammatory effect in SH-SY5Y cells.

## Future Work

Future work could explore whether Ashwagandha affects gene expression related to neuroinflammation and stress resilience through epigenetic mechanisms. Techniques like qPCR or RNA sequencing could help identify whether Ashwagandha alters the activity of key genes, such as those involved in cytokine production or stress regulation. This could provide insight into whether its effects are temporary or involve longer-lasting molecular changes.

## References

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- Soufli, I., Toumi, R., Rafa, H., & Touil-Boukoffa, C. (2016). Overview of cytokines and nitric oxide involvement in immunopathogenesis of inflammatory bowel diseases. *World Journal of Gastrointestinal Pharmacology and Therapeutics*, 7(3), 353. <https://doi.org/10.4292/wjgpt.v7.i3.353>